

PREFACE FOR FACULTY

While basic windows (Hann, Bartlett) provide moderate stopband attenuation, high-performance applications (audio equalization, communications, radar pulse compression) demand deep stopband rejection (> 50 dB). The **Hamming**, **Blackman**, and **Kaiser** windows provide superior stopband suppression.

Pedagogical Strategy:

- 1. Formulate the **Hamming Window**: $w[n] = 0.54 - 0.46 \cos(\frac{2\pi n}{N-1})$, showing how the 0.54/0.46 coefficient choice places a spectral zero directly on the first sidelobe peak, yielding -43 dB peak sidelobe.
- 2. Formulate the **Blackman Window**: Adding a 2nd harmonic cosine achieves -58 dB peak sidelobe and 74 dB attenuation.
- 3. Master the **Kaiser Window Family**: Parameterized by shape factor β and zeroth-order modified Bessel function $I_0(\cdot)$.
- 4. Teach empirical design formulas (Bellanger / Kaiser formulas) to calculate exact filter length N from transition width Δf and stopband attenuation A_s .
- 5. Complete comprehensive filter design examples.

1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

- 1. **Design** high-rejection FIR filters using Hamming, Blackman, and Kaiser windows.
- 2. **Compute** required filter length N using Kaiser empirical design equations.
- 3. **Tune** the shape parameter β to trade off transition bandwidth and stopband ripple.
- 4. **Compare** performance trade-offs across all standard window families.

2. MATHEMATICAL FOUNDATIONS

2.1 Standard Window Performance Summary

Window	Time Formula $w[n], ; 0 \leq n \leq N - 1$	Mainlobe Width $\Delta\omega$	Peak Sidelobe	Min Stopband Attenuation A_s	Transition Width Formula
Rectangular	1	$4\pi/N$	-13 dB	21 dB	$\Delta f \approx 0.9/N$
Hann	$0.5 - 0.5 \cos(\frac{2\pi n}{N-1})$	$8\pi/N$	-31 dB	44 dB	$\Delta f \approx 3.1/N$
Hamming	$0.54 - 0.46 \cos(\frac{2\pi n}{N-1})$	$8\pi/N$	-41 dB	53 dB	$\Delta f \approx 3.3/N$

Window	Time Formula $w[n]; 0 \leq n \leq N-1$	Mainlobe Width $\Delta\omega$	Peak Sidelobe	Min Stopband Attenuation A_s	Transition Width Formula
Blackman	$0.42 - 0.5 \cos\left(\frac{2\pi n}{N-1}\right) + 0.08 \cos\left(\frac{4\pi n}{N-1}\right)$	$12\pi/N$	-57 dB	74 dB	$\Delta f \approx 5.5/N$

2.2 The Kaiser Window Design Equations

$$w[n] = \frac{I_0\left(\beta \sqrt{1 - \left(\frac{n-\tau}{\tau}\right)^2}\right)}{I_0(\beta)}, \quad \tau = \frac{N-1}{2}$$

Where $I_0(x) = \sum_{k=0}^{\infty} \left[\frac{(x/2)^k}{k!} \right]^2$.

- **Shape Parameter β Calculation:**

$$\beta = \begin{cases} 0.1102(A_s - 8.7), & A_s > 50 \\ 0.5842(A_s - 21)^{0.4} + 0.07886(A_s - 21), & 21 \leq A_s \leq 50 \\ 0, & A_s < 21 \end{cases}$$

- **Filter Length N Calculation:**

$$N \geq \frac{A_s - 7.95}{14.36 \cdot \Delta f} + 1, \quad \Delta f = \frac{\omega_s - \omega_p}{2\pi}$$

3. WORKED NUMERICAL EXAMPLES

Example 23.1: Complete FIR Filter Design via Kaiser Window

Problem: Design a linear-phase FIR Lowpass filter meeting the following specifications:

- Passband edge: $\omega_p = 0.2\pi$ rad
- Stopband edge: $\omega_s = 0.3\pi$ rad
- Passband ripple: $\delta_p = 0.01$ (Peak passband ripple ≈ 0.086 dB)
- Stopband ripple: $\delta_s = 0.001$ (Stopband attenuation $A_s = -20 \log_{10}(0.001) = 60$ dB)

Solution:

1. **Determine Design Attenuation:** $A_s = -20 \log_{10}(\min(\delta_p, \delta_s)) = -20 \log_{10}(0.001) = 60$ dB.

2. **Compute Normalized Transition Bandwidth:**

$$\Delta f = \frac{\omega_s - \omega_p}{2\pi} = \frac{0.3\pi - 0.2\pi}{2\pi} = \frac{0.1\pi}{2\pi} = 0.05$$

3. **Calculate Filter Order & Length N :**

$$N \geq \frac{60 - 7.95}{14.36 \times 0.05} + 1 = \frac{52.05}{0.718} + 1 = 72.49 + 1 = 73.49 \implies \mathbf{N = 75} \quad (\text{Select odd length } N)$$

4. **Calculate Kaiser Parameter β ($A_s > 50$):**

$$\beta = 0.1102(60 - 8.7) = 0.1102(51.3) = \mathbf{5.6533}$$

5. **Cutoff Frequency:**

$$\omega_c = \frac{\omega_p + \omega_s}{2} = \frac{0.2\pi + 0.3\pi}{2} = 0.25\pi \text{ rad/sample}$$

4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

Question 1 (15 Marks)

(a) Explain why the Hamming window achieves -43 dB peak sidelobe level compared to -31 dB for the Hann window. (5 Marks) **(b)** Design an FIR bandpass filter using a Hamming window to pass frequencies between 0.3π and 0.6π with transition widths of 0.05π . Find N and the expression for $h[n]$. (10 Marks)

Model Answer & Step-by-Step Marking Rubric:

- **Part (a):**
 - Mathematical cancellation: The pedestal $0.54 - 0.46 \cos$ places a transmission zero at the first sidelobe peak of the Dirichlet sinc kernel (5 Marks)
- **Part (b):**
 - Transition width $\Delta\omega = 0.05\pi \implies \Delta f = 0.025$.
 - For Hamming: $\Delta\omega = \frac{6.6\pi}{N} \implies N \geq \frac{6.6}{0.05} = 132 \implies N = 133$ (odd length) (4 Marks)
 - Ideal Bandpass: $h_d[n] = \frac{\sin(0.6\pi(n-\tau))}{\pi(n-\tau)} - \frac{\sin(0.3\pi(n-\tau))}{\pi(n-\tau)}$, $\tau = 66$ (3 Marks)
 - Final coefficients $h[n] = h_d[n] \cdot [0.54 - 0.46 \cos(2\pi n/132)]$ (3 Marks)

5. PYTHON VERIFICATION SCRIPT

```
import scipy.signal as signal

# Kaiser filter design verification
N, beta = signal.kaiserord(60, 0.1) # 60 dB, dw = 0.1*pi (df = 0.05)
print(f"Kaiser length N = {N}, beta = {beta:.4f}")
```